Listing - Final Multi-Level Model

Fixed Effects:

- **Effect of the duration of the vowel on the average normalized F0:** $\text{mean\_z\_pitch} \sim \text{total\_duration}$
- **Effect of prevocalic consonant (/p/ vs /t/):** $+\text{C1}$
- **Effect of vowel category and interaction with C1 and duration of the vowel:** $+(\text{vowel} \times \text{C1} \times \text{total\_duration})$

Random Effects:

- **Random effect of child:** $+(1 \mid \text{name})$
- **Random effect of item nested within child:** $+(1 \mid \text{name/C1\_V})$

Full Model:

$$\text{mean\_z\_pitch} \sim (\text{total\_duration} + \text{C1}) * \text{vowel} + (1 \mid \text{name/C1\_V})$$

Model Comparison - Multi-Level Modelling Analysis

	Added term	Baseline model	Compared model	Npar	AIC	BIC	Loglik	Deviance	χ^2	df	p-value	Comparison output	Retained model
Null Model with random intercept per child(M1)	Duration fixed effect (M2)	M1	M2	4	8190.4	8215.3	-4091.2	8182.4	753.72	1	<0.001	Random intercept per vowel needed	M2
	Duration fixed effect (M3)	M2	M3	5	8052.4	8083.5	-4021.2	8042.4	140	1	<0.001	Duration fixed effect needed	M3
	C1 fixed effect (M4)	M2	M4	6	7956.0	7993.3	-3972.0	7944.0	98.454	1	<0.001	C1 fixed effect needed	M4
	V fixed effect (M5)	M4	M5	17	7449.6	7555.2	-3707.8	7415.6	528.38	11	<0.001	V fixed effect needed	M5
	C1 * V interaction (M6)	M5	M6	28	7414.7	7588.7	-3679.3	7358.7	56.925	11	<0.001	C1*V interaction needed	M6
	C1 * Duration interaction (M7)	M6	M7	29	7410.3	7590.5	-3676.1	7352.3	6.394	1	0.01145	C1 * Duration interaction needed	M7
	Duration * vowel interaction (M8)	M7	M8	51	7429.0	7745.9	-3663.5	7327.0	25.296	22	0.2831	Duration * vowel interaction not needed	M7
	Hearing status fixed effect (M9)	M6	M9	41	7428.7	7683.5	-3673.4	7346.7	0.2745	2	0.8718	Equivalent data fit	M6

Listing - Final generalised additive mixed model

Parametric term of C1 (i.e. fixed effect) on normalized pitch:

$$\text{final\_model} \leftarrow \text{bam}(\mathsf{z\_pitch} \sim \text{C1})$$

Single smooth (i.e. random intercept) for the effect of time_centred and total_duration:

$$+\mathsf{s}(\text{time\_centred}) + \mathsf{s}(\text{total\_duration})$$

Tensor product interaction of time_centred and total_duration (i.e. interaction between variables):

$$+\mathsf{ti}(\text{time\_centred}, \text{total\_duration})$$

Single smooth (i.e. random intercept) for the effect of child:

$$+\mathsf{s}(\text{name}, \text{bs} = \text{"}\mathsf{re}\text{"})$$

Difference smooth (i.e. random slope) for the effect of child per C1:

$$+\mathsf{s}(\text{name}, \text{C1}, \text{bs} = \text{"}\mathsf{re}\text{"})$$

Random smooth for the non-linear effect of time according to the general pattern of each speaker:

$$+\mathsf{s}(\text{time\_centred}, \text{name}, \text{bs} = \text{"}\mathsf{fs}\text{"}, \mathsf{m} = 1)$$

Tensor product interaction for the effect of child per C1 and hearing status:

$$+\mathsf{ti}(\text{time\_centred}, \text{total\_duration}, \text{by} = \text{C1})$$

$$+\mathsf{ti}(\text{time\_centred}, \text{total\_duration}, \text{by} = \text{Hearing status})$$

Model comparison - Generalised Additive Mixed Modelling

		Added term		Removed term	Baseline model	Compared model	Score	Edf	Difference	Df	p-value	Comparison output	Retained model
		Model single and random smooths per child (M1B)			M1	M1B	-66425.34	11	195341.267	3	<0.001	Single and random smooths per child needed	M1B
Null (M1)	Model	With C1 predictor (M2A)			M1B	M2A	-70447.85	23	4022.510	12	<0.001	C1predictorneeded	M2A
			With smooth term only (M2B)		M2A	M2B	-70447.85	23	14.500	1	<0.001	Parametric term needed	M2A
			With parametric term only (M2C)		M2A	M2C	-70447.85	23	3652.534	13	<0.001	Smooth terms needed	M2A
				Without tensor product interaction (M2A1)	M2A	M2A1	-70447.85	23	56.404	6	<0.001	Tensor product interaction needed	M2A
				Without single smooth (M2A2)	M2A	M2A2	-70449.01	22	-1.162	1	Only small difference in ML	Equivalent data fit	M2A2
				Without random smooth (M2A3)	M2A	M2A3	-70447.85	23	1923.328	4	<0.001	Random smooth needed	M2A
		With hearing status predictor (M3A)			M2A2	M3A	-70596.66	33	147.650	11	<0.001	Hearing status predictor needed	M3A
			With smooth term only (M3B)		M3A	M3B	-70572.56	35	0.458	2	0.633	Equivalent data fit	M3B
			With parametric term only (M3C)		M3A	M3C	-70572.56	35	124.263	11	<0.001	Smooth terms needed	M3A
				Smooth terms needed	M3B	M3C	-70572.1	33	123.806	9	<0.001		M3B
				Without tensor product interaction (M3B1)	M3B	M3B1	-70572.10	33	122.932	6	<0.001	Tensor product interaction needed	M3B
				Without single smooth (M3B2)	M3B	M3B2	-70572.95	32	-0.850	1	Only small difference in ML...	Equivalent data fit	M3B2
				Without random smooth (M3B3)	M3B	M3B3	-70572.11	29	-0.011	4	Only small difference in ML...	Equivalent data fit	M3B3
				Without random and single smooths (M3B4)	M3B4	M3B2	-70572.95	32	0.973	4	0.746	Equivalent data fit	M3B4
					M3B4	M3B3	-70572.11	29	0.134	1	0.604	Equivalent data fit	M3B4